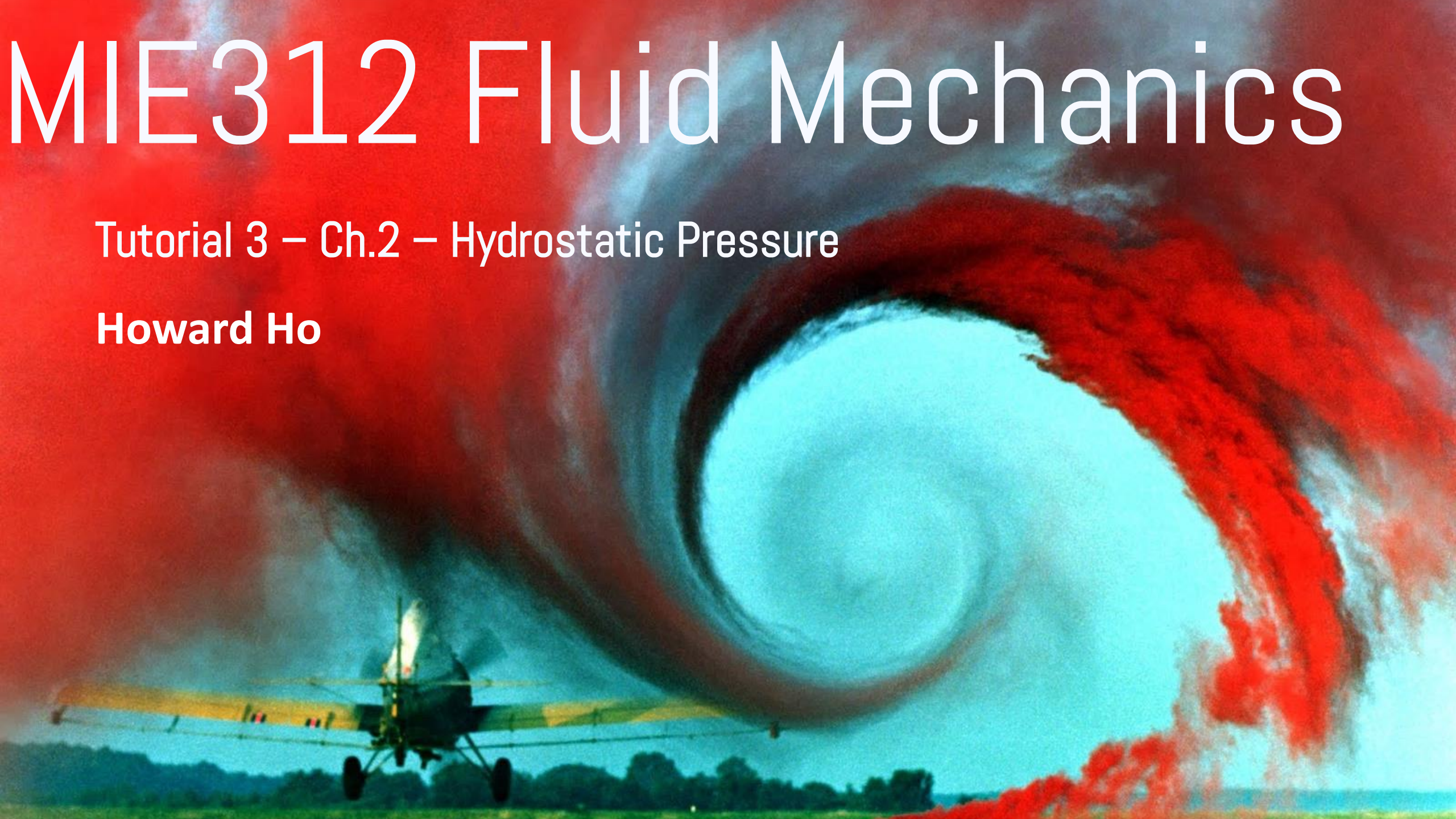
MIE312 Fluid Mechanics

Tutorial 3 – Ch.2 – Hydrostatic Pressure

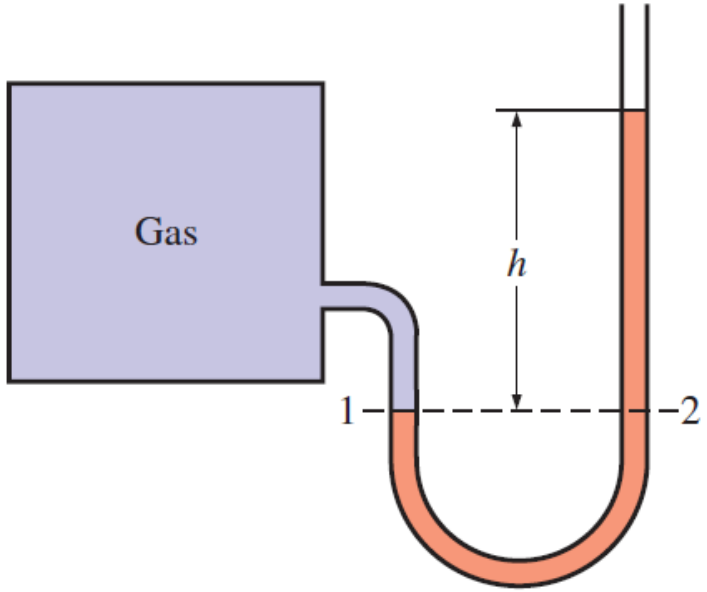
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Learning Objective

- Concept of pressure
- Calculate pressure using manometers
- Calculate the forces exerted by a fluid at rest on plane







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Hydrostatic Pressure



- Hydrostatic = fluids at rest
- Most medium-long haul airliners cruise at 30,000 feet
  - Air density is significantly lower and thus less aerodynamic drag
- Pressure equations
  - Gas:

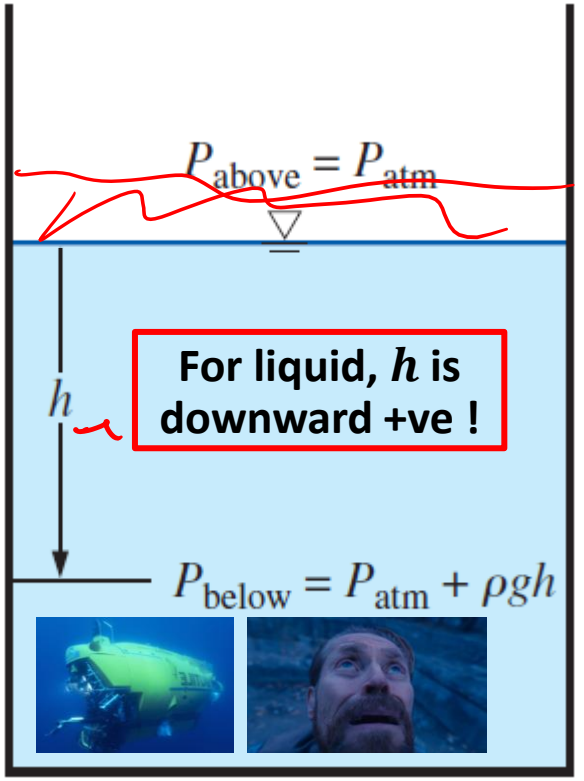
$$P = P_0\left(1 - \frac{\beta z}{T_0}\right)^{g/R\beta}$$

  - Liquids:

$$P = P_0 + \rho g h$$

$P_0 = Surface\ Pessure$

- Normally $P_0 = P_{atm}$, but only if liquid is exposed to atmosphere



$h = 3800\ m$ for titanic wreck
Comes out to 372 atm…

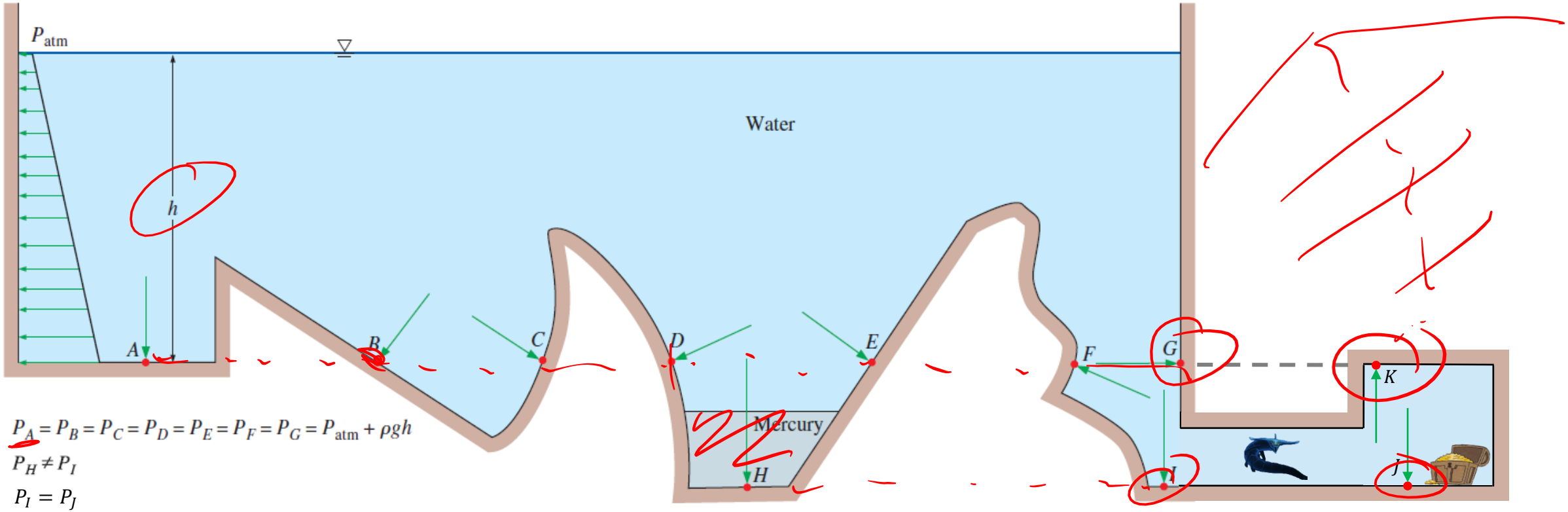
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Pressure

- Liquid Pressure Equation

$$P = P_o + \rho g h$$

- Pressure is same at the same depth, ignoring the seabed geometry (assuming same liquid)



$P_A = P_B = P_C = P_D = P_E = P_F = P_G = P_{atm} + \rho g h$

$P_H \neq P_I$

$P_I = P_J$

$P_G = P_K = P_{atm} + \rho g h$

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Pascal's Law

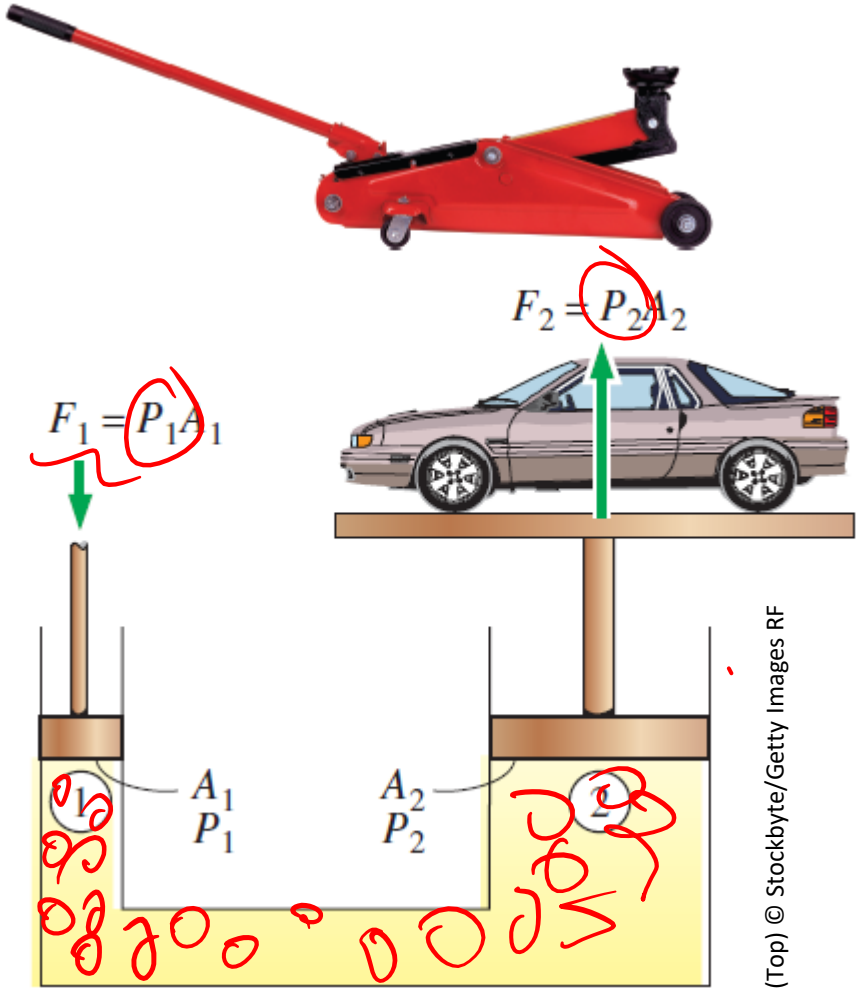
- A pressure change at any point in a confined **incompressible fluid** is transmitted throughout the fluid such that the same change occurs everywhere

$$F_{Pressure} = P \times A$$

$$P_1 = P_2 \quad \rightarrow \quad \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \rightarrow \quad \frac{F_2}{F_1} = \frac{A_2}{A_1}$$



Lifting of a large weight by a small force by the application of Pascal's law.



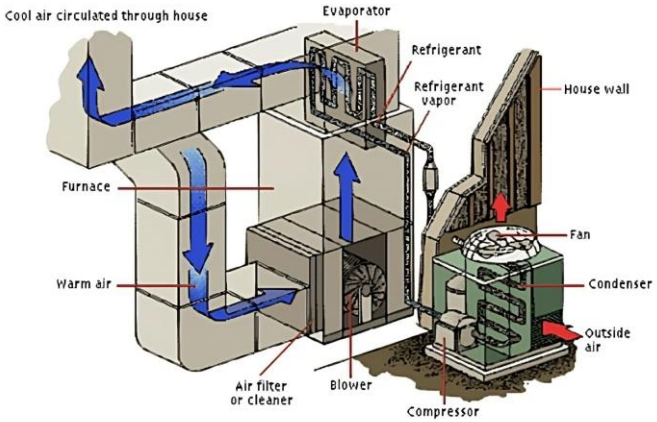
(Top) © Stockbyte/Getty Images RF

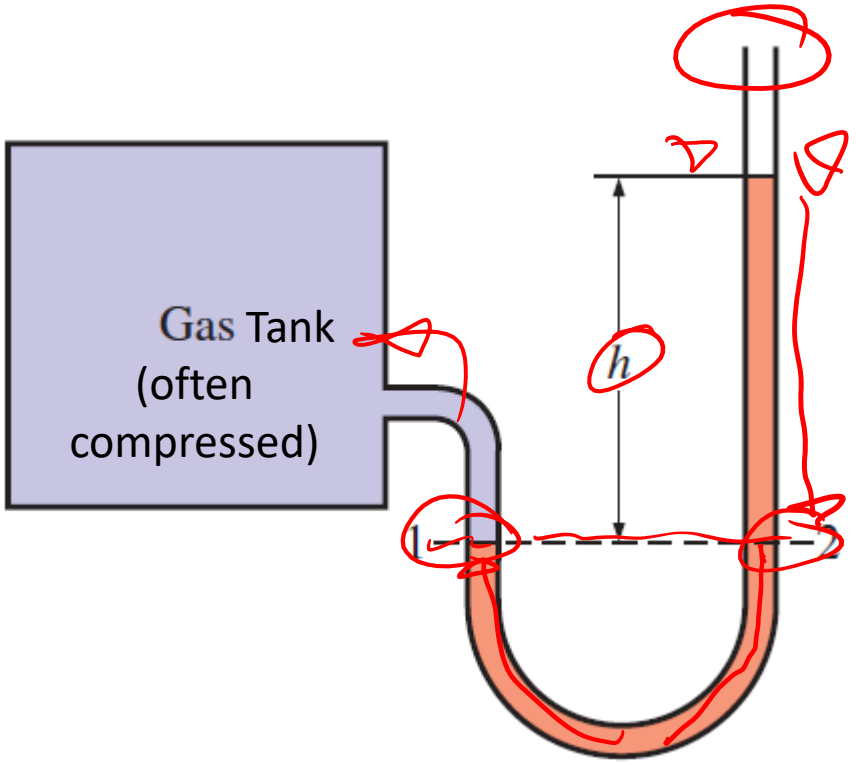
Manometer

- Measures small and moderate pressure differences.
- Broad applications in HVAC
- Three Pressure terms:

$$\underbrace{P_2}_{P_{absolute}} = \overbrace{P_{atm}}^{P_{atmospheric}} + \underbrace{\rho g h}_{P_{gauge}}$$

- $P_{gage} = P_{absolute} - P_{atm}$
- Think of P_{gage} as a "relative" pressure







Further explanation:
https://www.youtube.com/watch?v=gxrkLkJybnA

Cengel, Y. A. (2010). *Fluid mechanics*. Tata McGraw-Hill Education.

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Question Time

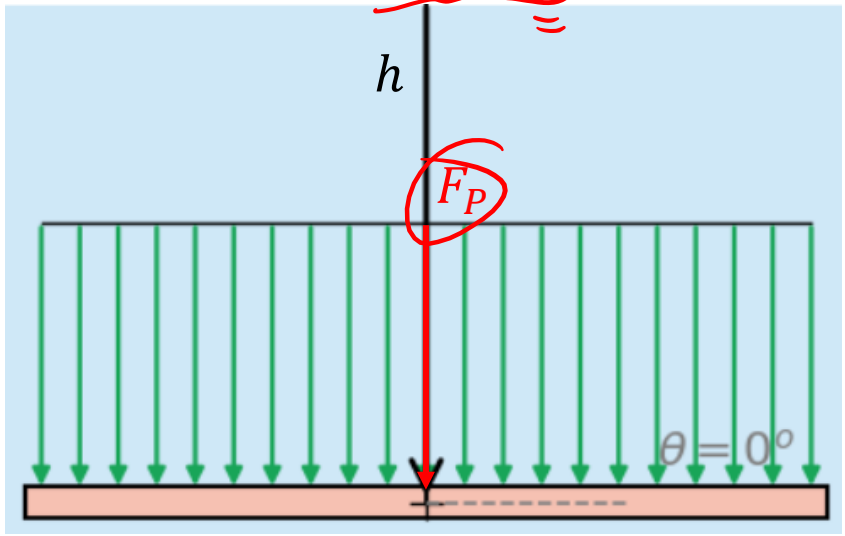


Hydrostatics-Plane Surface

- Pressure occurs on both sides of the plate if fully submerged
- When a plate is angled, pressure is not uniform over the surface
- This can result in a torque (if $F_{upper} \neq F_{lower}$)
- Need to adjust where pressure force F_P is acted on

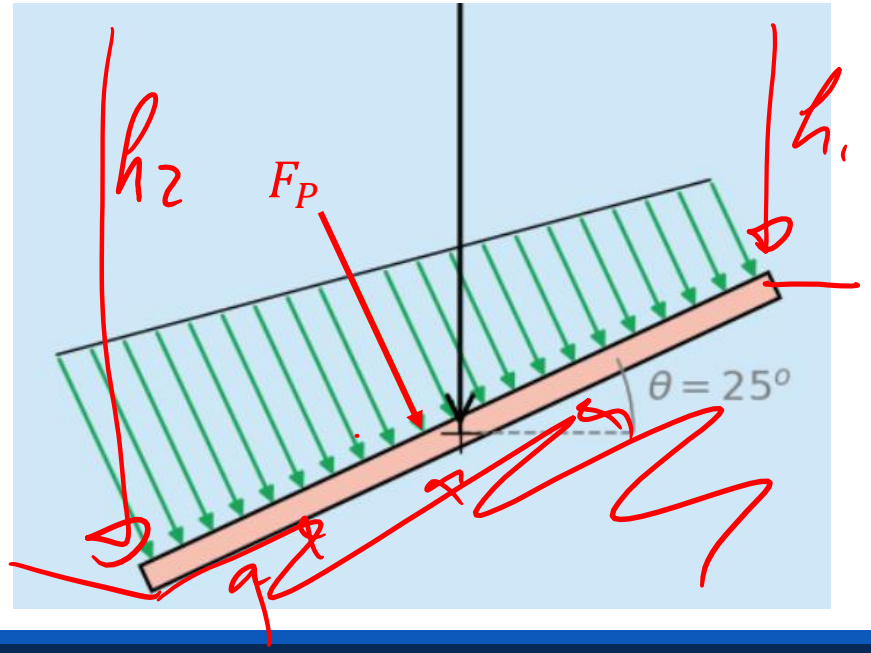
The magnitude of F_P is same no matter the θ, but the location that F_P acts on changes with θ

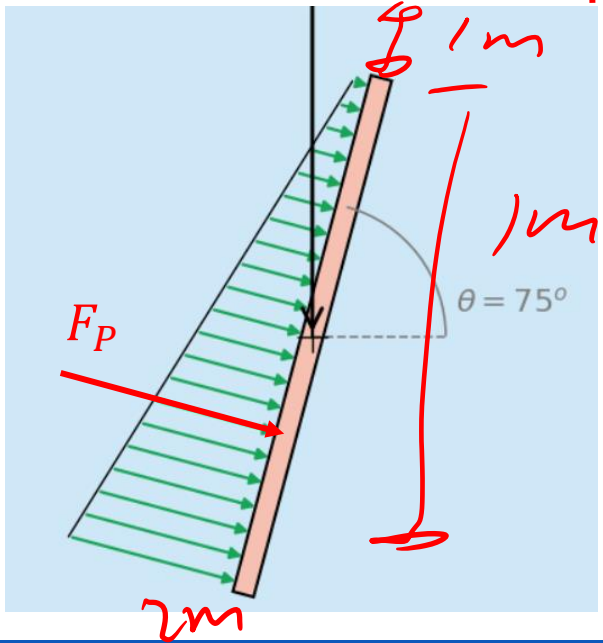
$$F_{gage} = P_{gage}\, A_{plate}$$
$$F_{gage} = \rho g h\, A_{plate}$$
$$F_{aboslute} = P_{atm} A_{plate} + \rho g h\, A_{plate}$$



Green arrows = pressure distribution

Need a bit more work to find where F_P acts on

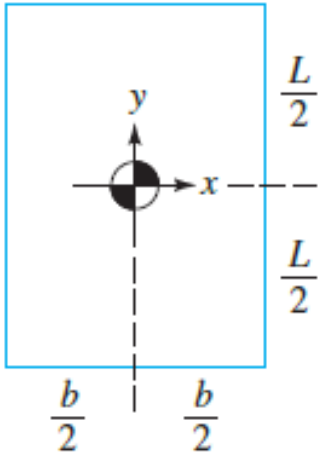




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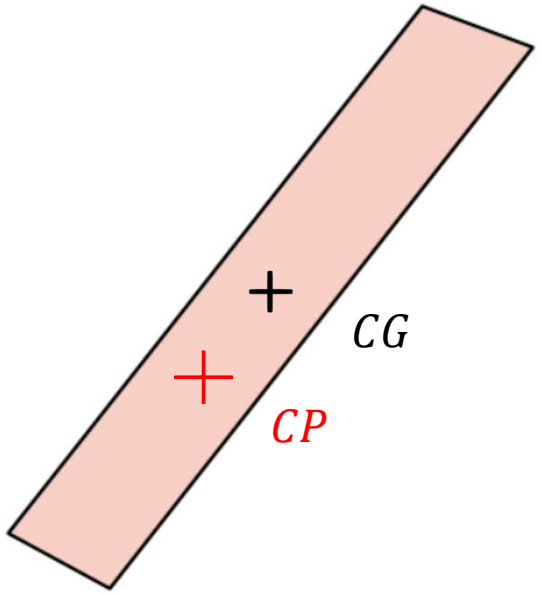
Angled Plates

- Pressure force F_P no longer act through Centre of Gravity (CG)
- Centre of Pressure (CP) = location of pressure force F_P
- y_{CP} is distance from CG to CP
- As θ increases, y_{CP} also increases

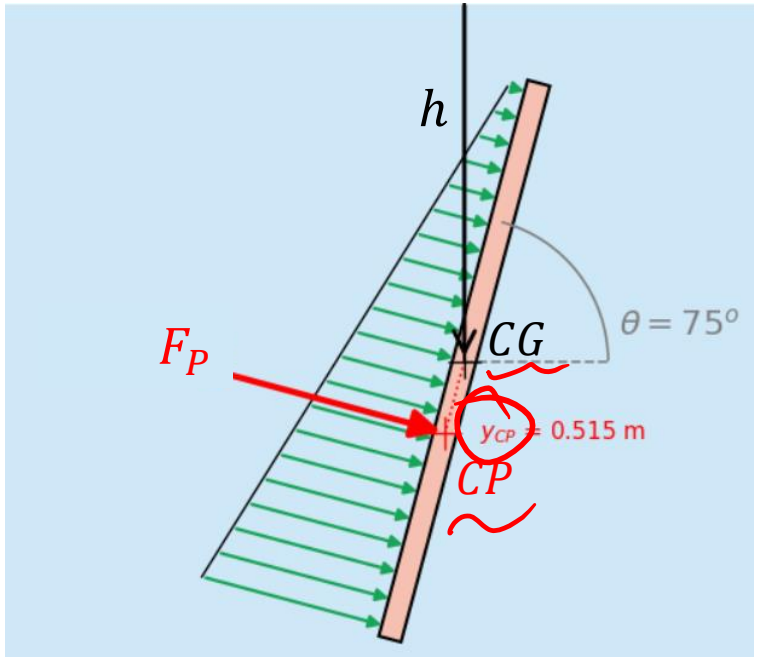


$$A = bL \qquad I_{xx} = \frac{bL^3}{12} \qquad I_{xy} = 0$$

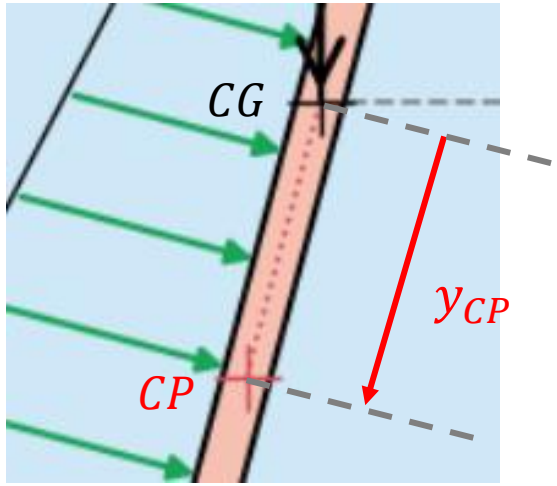
Isometric view



Side View



$$F = \gamma h_{CG} A \qquad y_{CP} = +\frac{I_{xx} \sin\theta}{h_{CG} A} \qquad x_{CP} = -\frac{I_{xy} \sin\theta}{h_{CG} A}$$



The y axis is not vertical, y axis is parallel to the plate !



Important Note: The textbook y_{CP} assumes y as upward +ve, we omit the –ve sign for y_{CP} with this coordination system.

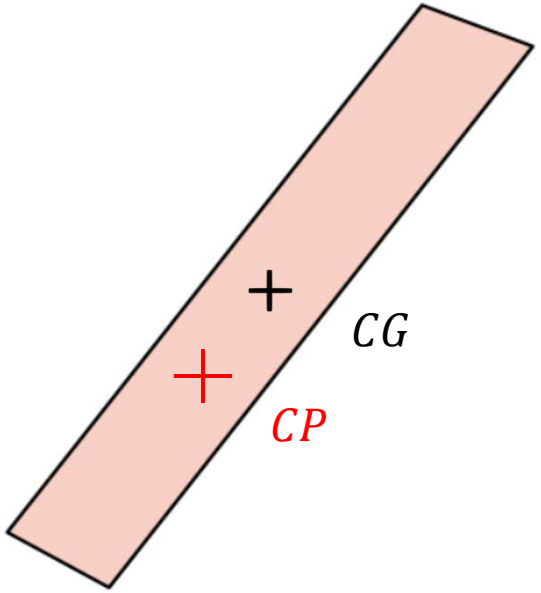
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3 Steps Guide*

- Step 1: Determine magnitude of the force using Centre of Gravity (CG)

$$F_P = P \times A$$

$$F_P = \rho g h_{CG} \times A$$



- Step 2: Find the point of pressure force - the Centre of Pressure (CP)

$$y_{cp} = + \frac{I_{xx} \sin\theta}{h_{CG} A}$$

- Step 3 (optional): If asked to find depth of CP

2. HYDROSTATICS & ATMOSPHERE

Statics: $P_2 - P_1 = -\gamma(z_2 - z_1)$; **HGL:** $HGL = z + P/\gamma$
Isothermal: $P = P_1 \exp[-g\Delta z/RT]$
Troposphere: $P = P_0(1 - \beta z/T_0)^{g/R\beta}$ ($\beta = 0.0065$ K/m)
Hydrostatic: $P = P_0 + \rho gh$
Pascal Law: $F = P \times A$; $P_2 = P_1$
Planes: $F = \gamma h_{CG} A$; $y_{CP} = \frac{I_{xx}\sin\theta}{h_{CG}A}$
Curved: $F_H = \gamma h_{CG} A_{proj}$; F_V = weight of fluid above
Buoyancy: $F_B = \rho g \forall_{sub}$
Stability: $\overline{MG} = \frac{I_o}{\forall_{sub}} - \overline{GB}$ (Stable if $\overline{MG} > 0$)
Rigid Body: $\nabla P = \rho(\vec{g} - \vec{a})$; **Rotate:** $z = \frac{\omega^2 r^2}{2g} + C$
Moving manometer: $A_1 a = A_2 b$; inclined-leg vertical change $= b\sin\theta$; write initial/final pressure walks and subtract.

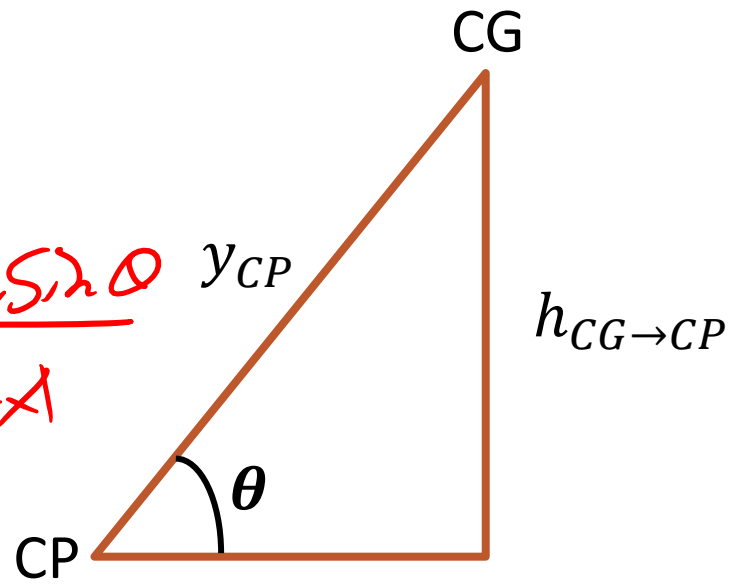
$$\sin\theta = \frac{h_{CG \to CP}}{y_{CP}}$$

$$h_{CP} = h_{CG} + h_{CG \to CP}$$

$I_{xx}\sin\theta$

$y_{CP} = y_{CG} + h_{CG}A$

Old crib sheet had a typo, please update yours to $y_{CP} = \frac{I_{xx}\sin\theta}{h_{CG}A}$



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Question Time



Problem 1

The water in a tank is pressurized by air, and the pressure is measured by a multifluid manometer as shown in Fig. P3–17. Determine the gage pressure of air in the tank if $h_1 = 0.4$ m, $h_2 = 0.6$ m, and $h_3 = 0.8$ m. Take the densities of water, oil, and mercury to be 1000 kg/m^3, 850 kg/m^3, and 13, 600 kg/m^3, respectively.

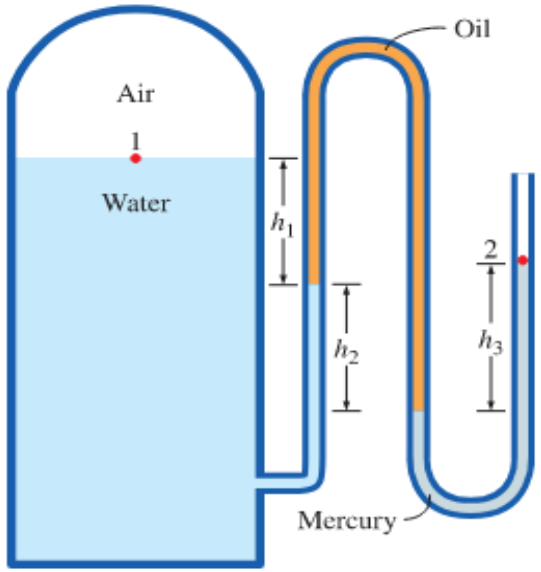


FIGURE P3–17

$$P_{gage} = P_{asb} - P_{atm}$$

2. HYDROSTATICS & ATMOSPHERE

Statics: $P_2 - P_1 = -\gamma(z_2 - z_1)$; **HGL:** $HGL = z + P/\gamma$
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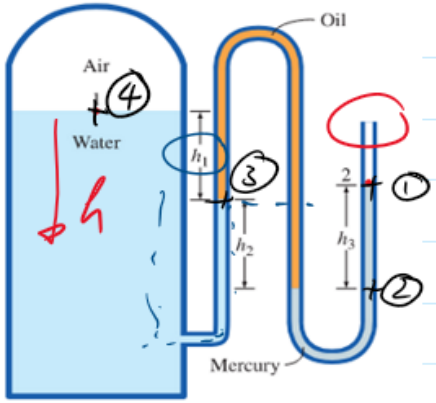
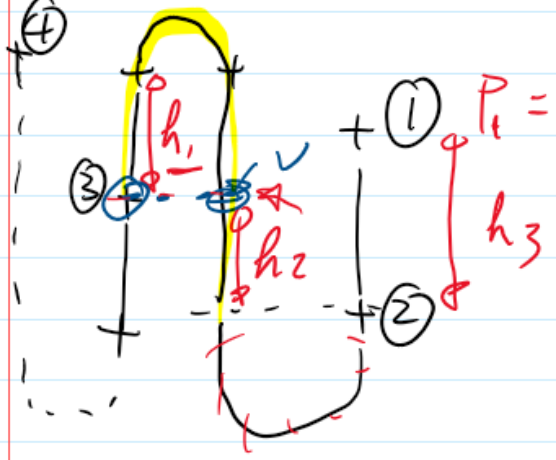


FIGURE P3–17



$P_1 = P_{atm}$

Start from P_{atm}

$P_1 \to P_2 = P_{atm} + \rho_w g h_3$

$P_2 = 100(1000) + 13,600(9.81)(0.8)$

$P_{gage} = P_{asb} - P_{atm}$

$P_2 = 206,732$ Pa

2. HYDROSTATICS & ATMOSPHERE

Statics: $P_2 - P_1 = -\gamma(z_2 - z_1)$; **HGL:** $HGL = z + P/\gamma$
Isothermal: $P = P_1 \exp[-g\Delta z/RT]$
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$P_2 \to P_3 = P_2 + \rho_{oil} g (-h_2)$ ~~$(-h_2 - h_1 + h_1)$~~

$P_3 = 206,732 + 850(9.81)(-0.6)$

$P_3 = 201,729$ Pa

$P_3 \to P_4 = P_3 + \rho_{water} g (-h_1)$

$P_4 = 197,805$ Pa

$P_{4,gage} = 97,805$ Pa

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Problem 2*

3–50 The pressure difference between an oil pipe and water pipe is measured by a double-fluid manometer, as shown in Fig. P3–50. For the given fluid heights and specific gravities, calculate the pressure difference $\Delta P = P_B - P_A$.

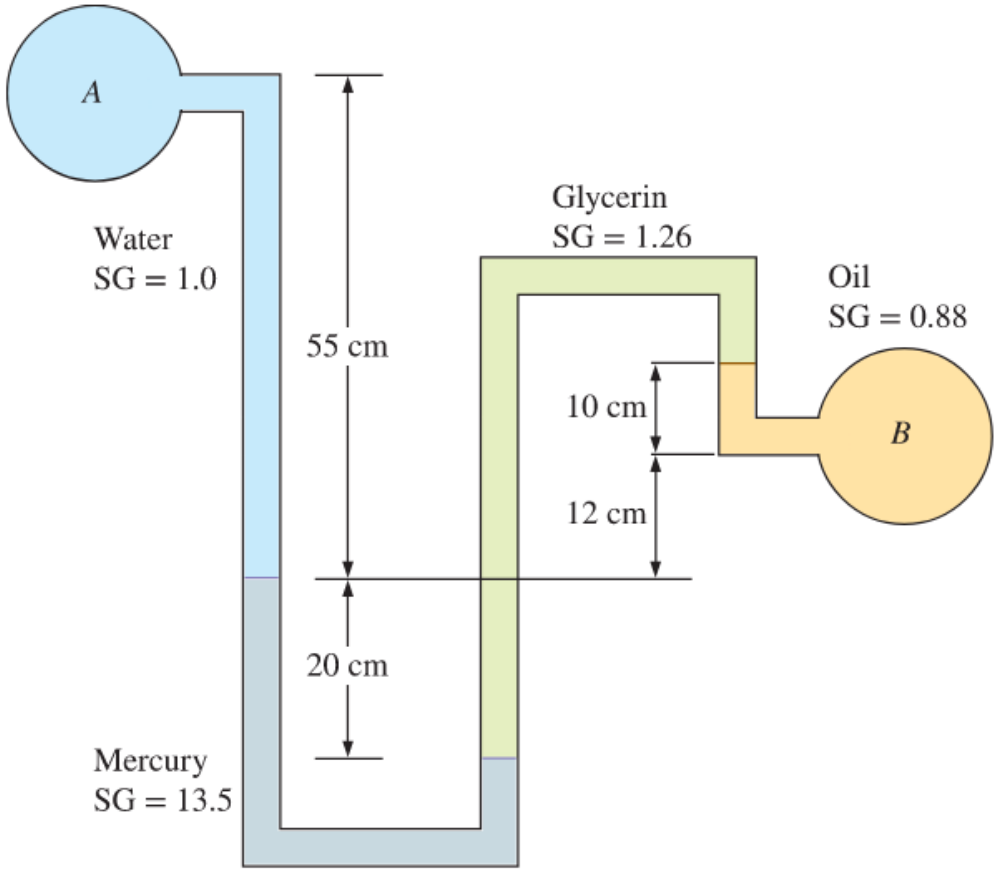


FIGURE P3–50

$$\rho_w = 1000\frac{kg}{m^3}, \rho_m = 13500, \rho_G = 1260\,, \rho_o = 880$$

starting from point A we add the pressures if we go down, skip horizontally, and subtract pressure when we go up .

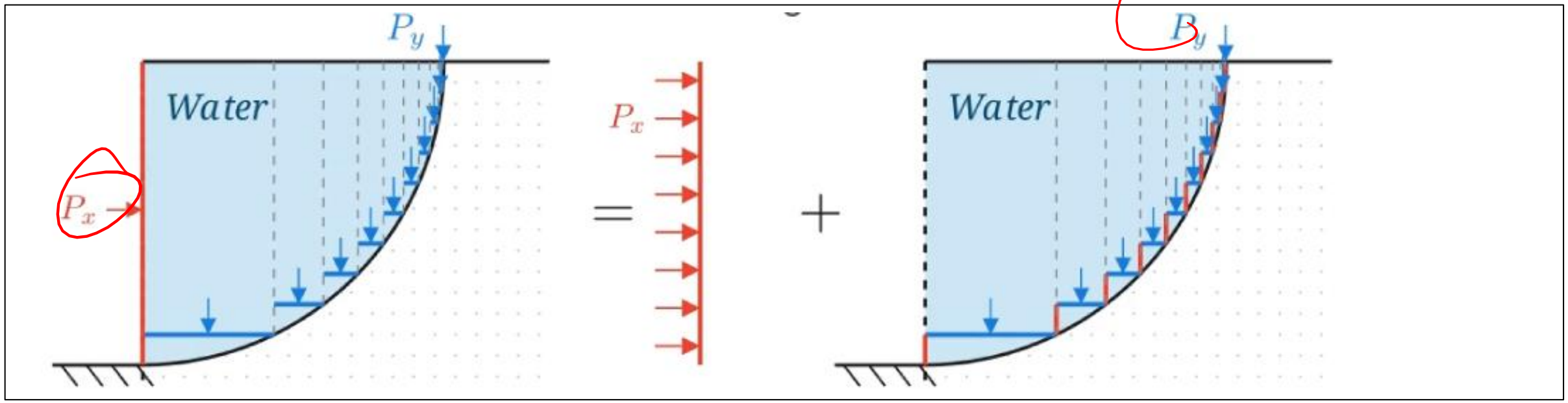
$$P_A + \rho_w g(0.55m) + \rho_m g(0.2m) - \rho_G g(0.2m + 0.12 + 0.1) + \rho_O g(0.1m) = P_B$$

$$P_B - P_A = \rho_w g(0.55m) + \rho_m g(0.2m) - \rho_G g(0.2m + 0.12 + 0.1) + \rho_O g(0.1m) = 27.6\,KPa$$

Therefore the oil is 27.6 KPA higher than the water

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Pressure on Curved Surface



- For F_x, we treat it as a vertical plate:

$$F_x = P_x \times A_{vertical}$$
$$= \rho g h_{CG} A_{vertical}$$

- For F_y, we use the weight of water above:

$$F_y = W$$
$$F_y = \rho g \forall_{above}$$

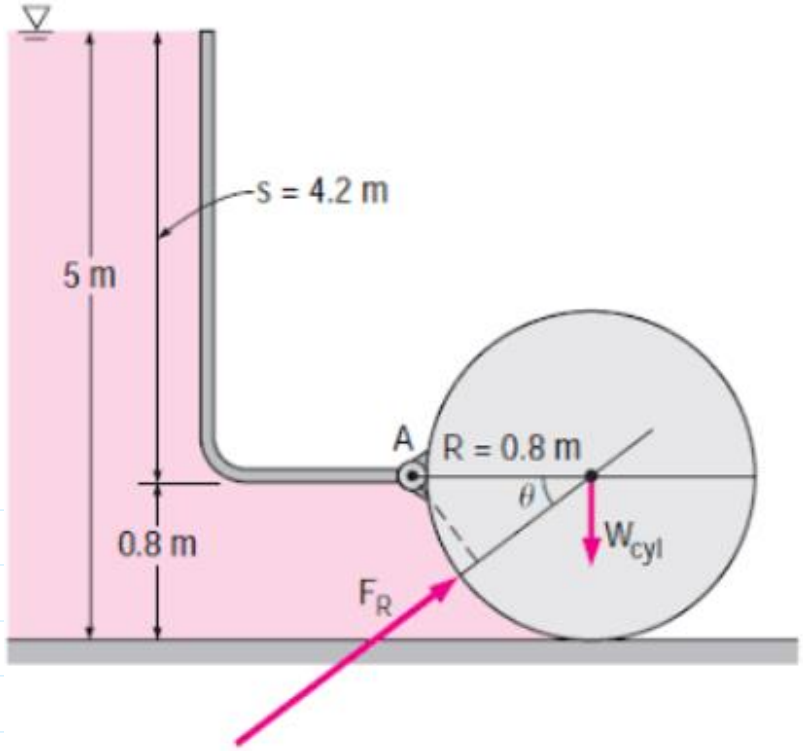
Important note, F_x does not act at h_{CG} it acts at h_{CP}

Link to animations:
https://howardho-cfd.github.io/teaching/fluids/interactive-tools/curved-plate-pressure/

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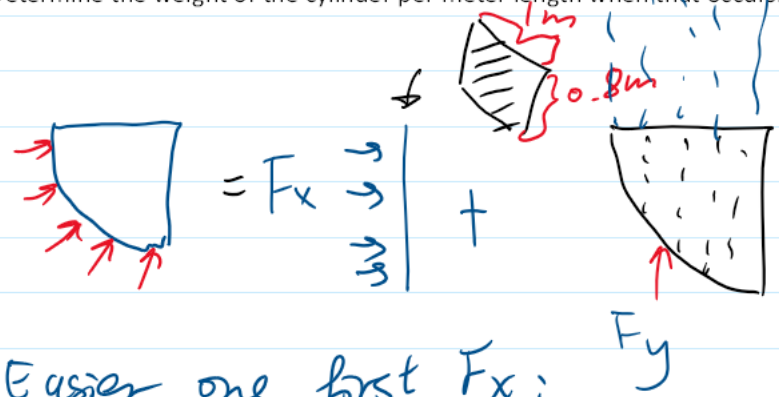
Question 3

A long solid circular cylinder of radius 0.8 m hinged at point A is used as an automatic gate. When the water level reaches 5 m, the gate opens *"slightly"* to release water, Determine the weight of the cylinder per meter length when that occurs.



October 1, 2026 5:47 PM

A long solid circular cylinder of radius 0.8 m hinged at point A is used as an automatic gate. When the water level reaches 5 m, the gate opens *"slightly"* to release water, Determine the weight of the cylinder per meter length when that occurs.

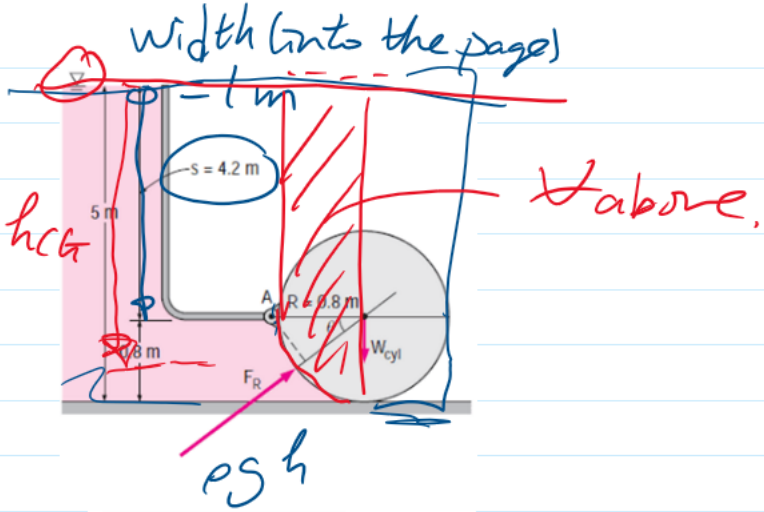


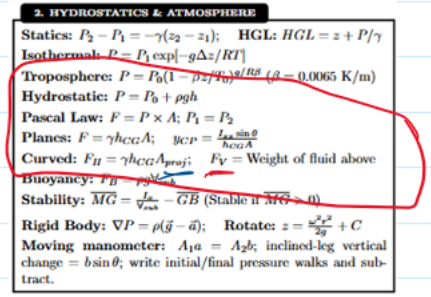
Easier one first F_x:

$F_x = \rho g h_{CG} A_{vertical}$

$F_x = 1000(9.81)(5-0.4)(0.8 \times 1)$

$F_x = 36101\ N$





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Isothermal: $P = P_1 \exp[-g\Delta z/RT]$
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The harder F_y: of volume $\forall$

$F_y = \rho g \forall_{above}$

$F_y = 1000(9.81)\left[(4.2 \times 0.8) + (\pi(0.8)^2/4)\right]$

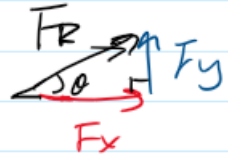
$F_y = 37893\ N$

Resultant from Pressure

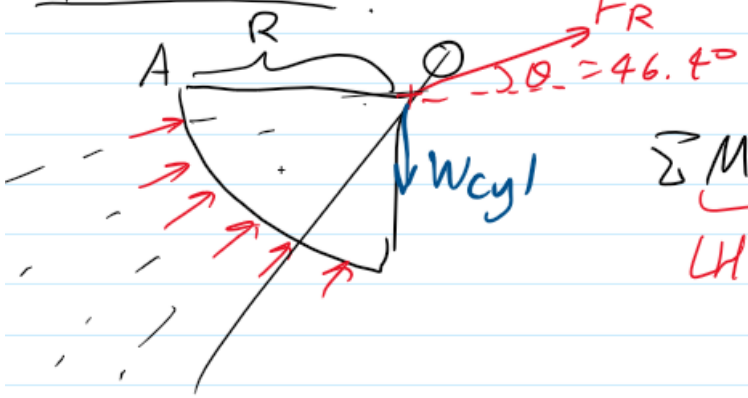
$F_R = \sqrt{F_x^2 + F_y^2}$ $\quad \tan\theta = \frac{F_y}{F_x}$

$= 52337\ N$ $\quad = \frac{37893}{36101}$

$\theta = 46.4°$



Update FBD



F_R acts through origin

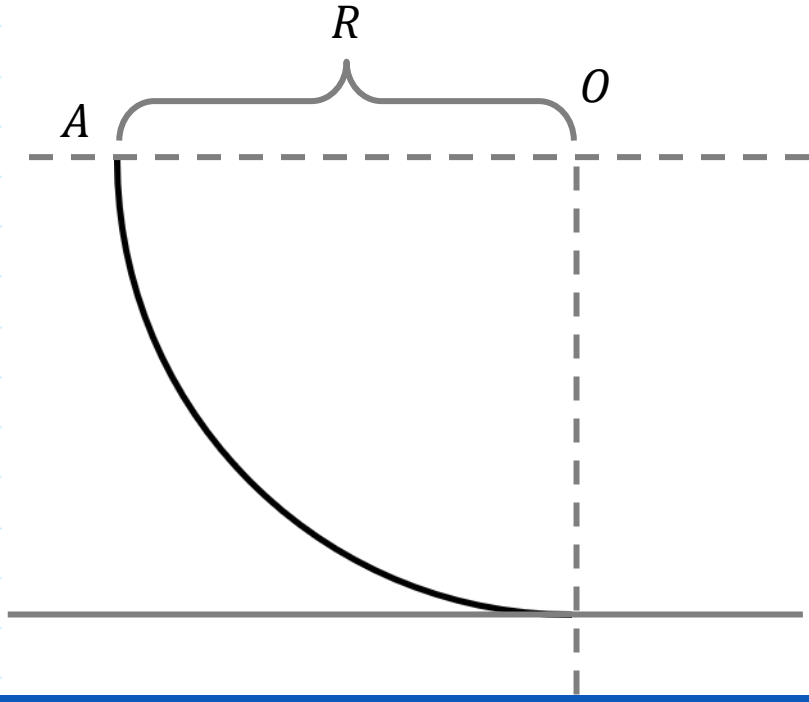
F_y

$\sum M_A\ (CCW) = R W_{cyl} - R F_R \sin\theta$

LHS = 0

$R W_{cyl} = R F_y$

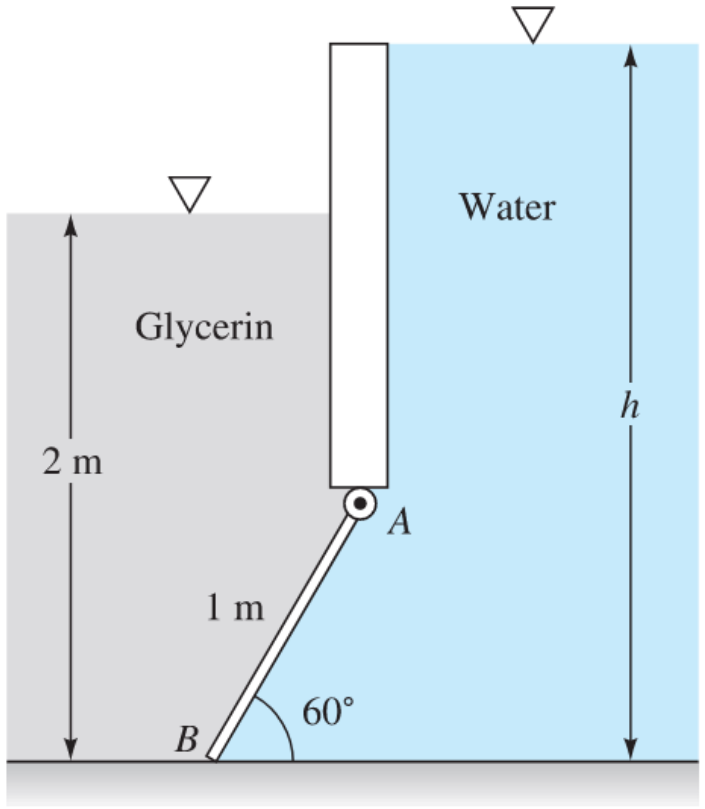
$\therefore W_{cyl} = 37893\ N$



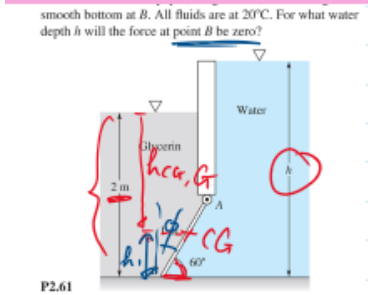
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Problem 4* (From 2024)

***P2.61** Gate AB in Fig. P2.61 is a homogeneous mass of 180 kg, 1.2 m wide into the paper, hinged at A, and resting on a smooth bottom at B. All fluids are at 20°C. For what water depth h will the force at point B be zero?



P2.61



FBD. $\Sigma M(A) = 0$

$\rho_{Gly} \cdot g = 12360\ N/m^3 = \gamma_G$

$\rho_w \cdot g = 9790\ N/m^3 = \gamma_w$

Find CG.

$h_1 = \frac{\sin 60^\circ}{2} = 0.433\ m$

$h_{CG,G} = 2 - h_1 = 1.567\ m$

Glycerin: $F = P \cdot A$

$F_G = \rho_G \cdot g \cdot h_{CG,G} \cdot A$

$= 12360\ (1.567)\ (1.2)$

$= 23242\ N$

$y_{CP,G} = -\frac{I_{xx} \sin\theta}{h_{CG,G} A}$

$= -\frac{0.1 \sin 60^\circ}{1.567\ (1.2)}$

$= -0.046\ m$

Water: $F_w = \rho_w \cdot g \cdot h_{CG,w} \cdot A$

$F_w = 9790\ (1.2)\ (h_{CG,w})$ — (1)

$y_{CP,w} = -\frac{I_{xx} \sin\theta}{h_{CG,w} A}$

$= -\frac{0.072}{h_{CG,w}}$ — (2)

Weight: $W = 180\ (9.81)$

$= 1766\ N$

$\Sigma M(A) = 0$

$F_G (y_G + y_{CP,G}) + 0.5 \cos 60^\circ (W) = F_w (y_G + y_{CP,w})$

$23242\ (0.5 + 0.046) + 441.5$

$= 11748\ (h_{CG,w}) \left[0.5 + \left(\frac{0.072}{h_{CG,w}}\right)\right]$

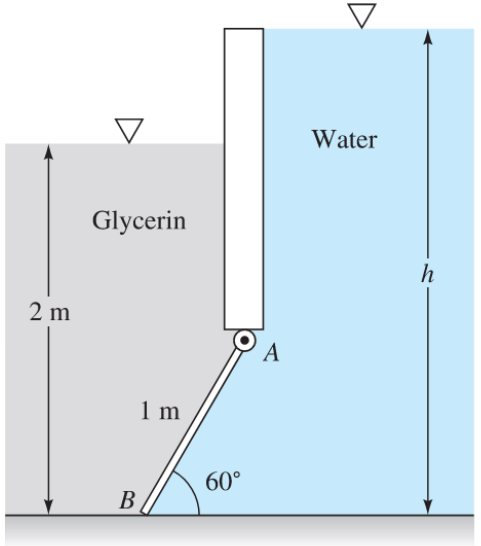
$h_{CG,w} = 2.09\ m$

$h = h_{CG,w} + h_1$

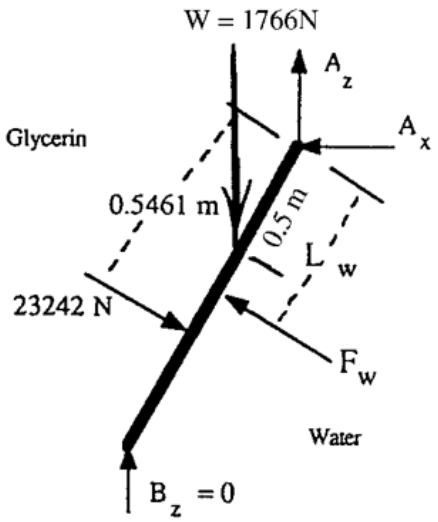
$= 2.52\ m$

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Problem 4*



***P2.61** Gate *AB* in Fig. P2.61 is a homogeneous mass of 180 kg, 1.2 m wide into the paper, hinged at *A*, and resting on a smooth bottom at *B*. All fluids are at 20°C. For what water depth *h* will the force at point *B* be zero?



Solution: Let $\gamma = 12360$ N/m^3 for glycerin and 9790 N/m^3 for water. The centroid of AB is 0.433 m vertically below A, so $h_{CG} = P2.0 - 0.433 = 1.567$ m, and we may compute the glycerin force and its line of action:

$$F_g = \gamma \bar{h} A = (12360)(1.567)(1.2) = 23242 \text{ N}$$

$$y_{CP,g} = -\frac{(1/12)(1.2)(1)^3 \sin 60°}{(1.567)(1.2)} = -0.0461 \text{ m}$$

These are shown on the freebody below. The water force and its line of action are shown without numbers, because they depend upon the centroidal depth on the water side:

$$F_w = (9790) h_{CG} (1.2)$$

$$y_{CP} = -\frac{(1/12)(1.2)(1)^3 \sin 60°}{h_{CG}(1.2)} = -\frac{0.0722}{h_{CG}}$$

The weight of the gate, W = 180(9.81) = 1766 N, acts at the centroid, as shown above. Since the x-force at B equals zero, we may sum moments counterclockwise about A to find the water depth:

$$\Sigma M_A = 0 = (23242)(0.5461) + (1766)(0.5 \cos 60°) - (9790) h_{CG} (1.2)(0.5 + 0.0722/h_{CG})$$

Solve for $h_{CG,water} = 2.09$ m, or: $h = h_{CG} + 0.433 = \mathbf{2.52\ m}$ *Ans.*

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Method 1 (F_R based)

- Step 1: Find F_x and F_y
- Step 2: Calculate resultant pressure force F_R and θ

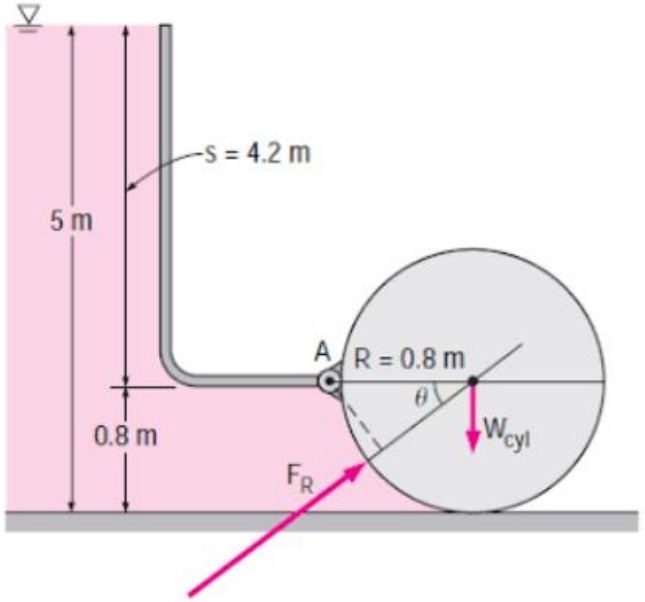
$$F_R = \sqrt{F_x^2 + F_y^2} \qquad \& \qquad \tan\theta = \frac{F_y}{F_x}$$

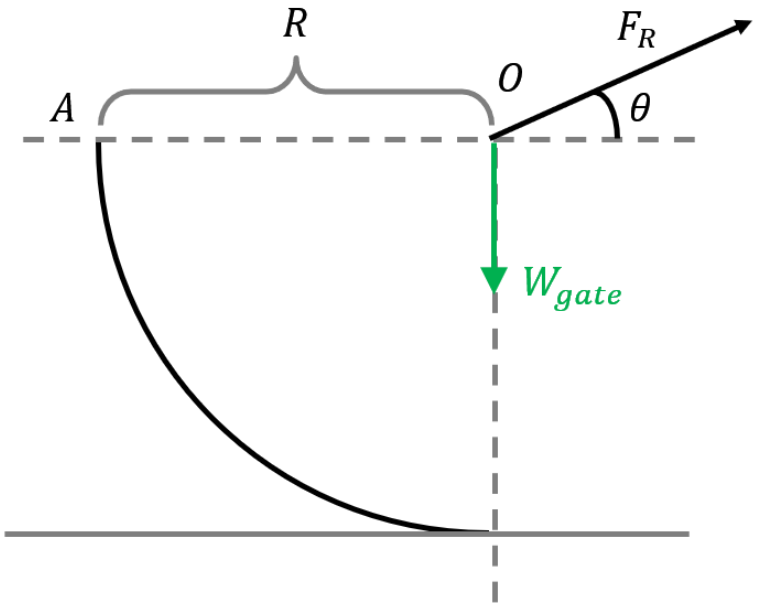
- Step 3: Take moment at pivot point

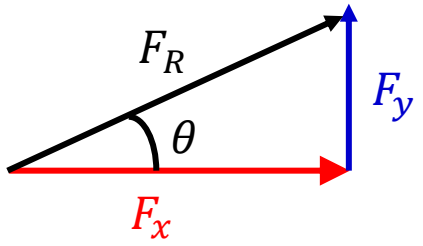
$$\Sigma M_B(CW) = R \times W_{gate} - R \times F_R \sin\theta$$

- Gate is not moving, therefore $\Sigma M_B = 0$, momentum equation becomes:

$$W_{gate} = F_R \sin\theta$$







Fun fact, $F_R \sin\theta = F_y$

All we did was a fancy $W_{gate} = F_y$

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Method 2 (F_x and F_y based)

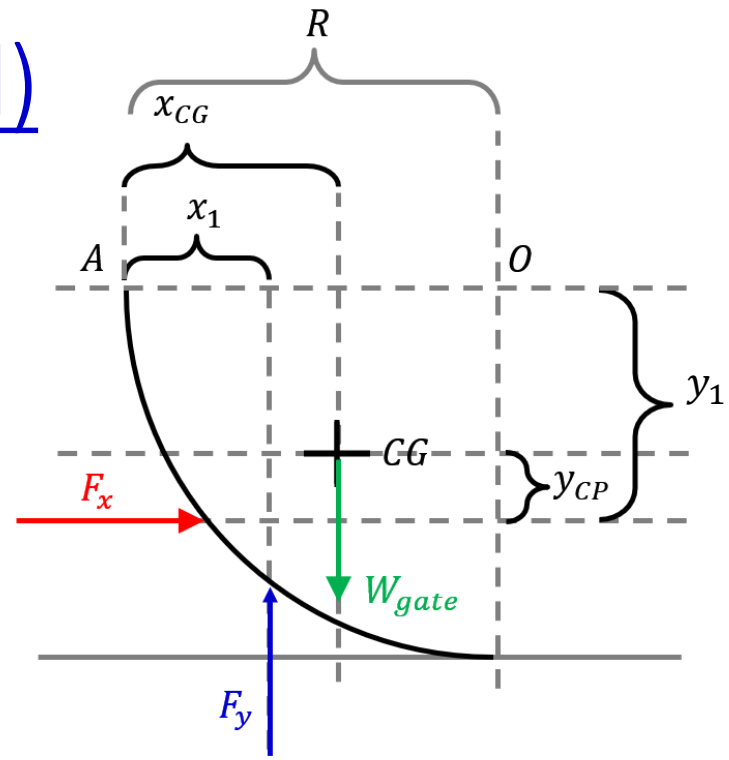
- Step 1: Find F_x and F_y
- Step 2: Find where F_x and F_y acts

$$y_{cp} = +\frac{I_{xx}\sin\theta}{h_{CG}A}, \quad \sin\theta = \frac{h_{CG\to CP}}{y_{CP}} \quad \& \quad h_{CP} = h_{CG} + h_{CG\to CP}$$

- y_1 is where F_x acts

$$y_1 = h_{CP} - 4.2$$

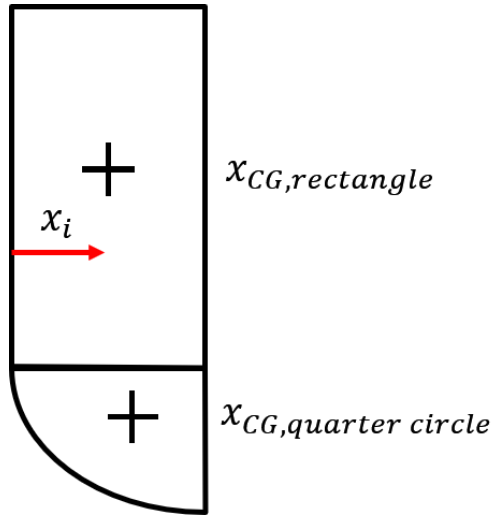
- x_1 is the centroid of the virtual volume above (rectangle + quarter circle), it's where F_y acts



Centroid of quarter circle $x_{CG,qc} = \frac{4r}{3\pi}$

$$x_1 = \frac{\Sigma x_i A_i}{\Sigma A_i}$$

x_i is distance to centroid of each sub geometry (starting from the left side here)



$$x_1 = \frac{0.4\times(4.2\times0.8) + \left(0.8 - \frac{4r}{3\pi}\right)\times\left(\pi\times\frac{r^2}{4}\right)}{(4.2\times0.8) + (\pi\times\frac{r^2}{4})}$$

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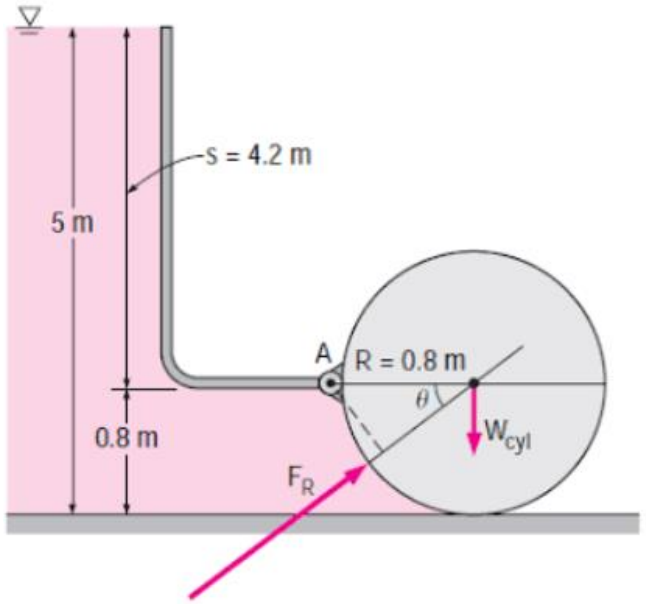
Method 2 (F_x and F_y based)

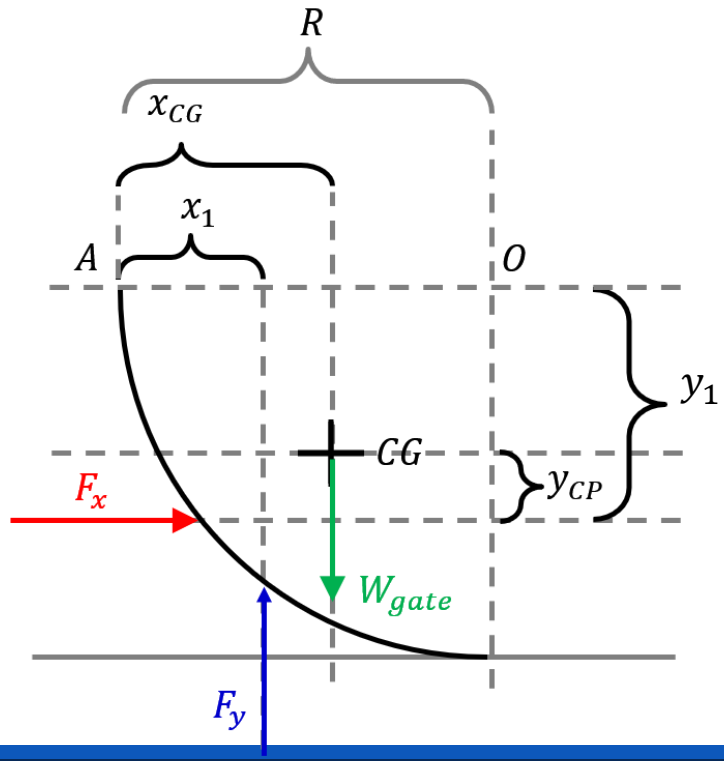
- Step3: Take moment at pivot point A

$$\Sigma M_A(CW) = R \times W_{gate} - x_1 \times F_y - y_1 \times F_x$$

- Gate is not moving, therefore $\Sigma M_A = 0$, momentum equation becomes:

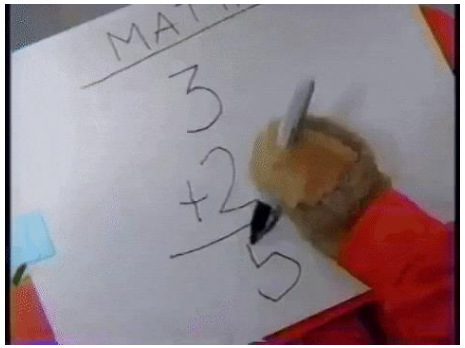
$$x_1 \times F_y + y_1 \times F_x = R \times W_{gate}$$





Method 2 is about 2-3 times longer than method 1, there's also a lot more room for error.

"We can do this the easy way, or we can do it the hard way. The choice is up to you."





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Problem 1

3–17 The water in a tank is pressurized by air, and the pressure is measured by a multifluid manometer as shown in Fig. P3–17. Determine the gage pressure of air in the tank if $h_1 = 0.4$ m, $h_2 = 0.6$ m, and $h_3 = 0.8$ m. Take the densities of water, oil, and mercury to be 1000 kg/m^3, 850 kg/m^3, and 13,600 kg/m^3, respectively.

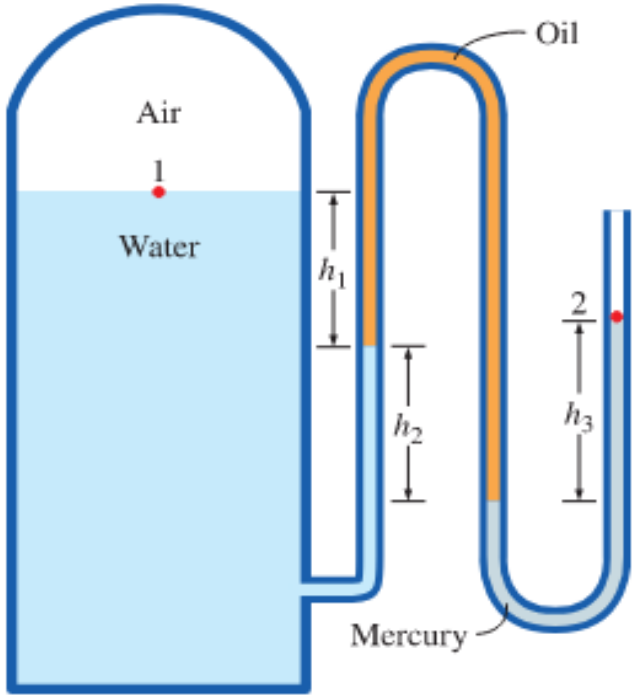


FIGURE P3–17

Work your way from point 1 to point 2 :

$$P_1 + \rho_w g h_1 + \rho_o g h_2 - \rho_m g h_3 = P_2$$

Solve for $P_1 - P_2$ which is $P_{1_{Gauge}}$,

$$P_1 = -\rho_w g h_1 - \rho_o g h_2 + \rho_m g h_3 + P_{atm/pt2}$$

$$P_1 - P_2 = P_{1gauge} = -\rho_w g h_1 - \rho_o g h_2 + \rho_m g h_3 = g(-\rho_w h_1 - \rho_o h_2 + \rho_m h_3)$$

$$P_1 - P_2 = P_{1_{Gauge}} = 97.8\ \text{KPa}$$

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Feedback Form & Office Hour

- Note: this feedback form is only for sessions run by Howard Ho, please only fill it out **after** your session of tutorial 3.
- Q5 of the feedback form allows me to pick the time for the office hour (which will be announced on Friday 6pm so plz fill this up asap)

5. For the Mid term 1 office hour, which of the following works for you?

- [ ] Tuesday (Oct-6th) 2-4pm
- [ ] Tuesday (Oct-6th) 4-6pm
- [ ] Wednesday (Oct-7th) 10am - 12pm
- [ ] Wednesday (Oct-7th) 12pm - 2pm
- [ ] Wednesday (Oct-7th) 4pm - 6pm

MIE 312 - Tutorial Feedback - 2026

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